

PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL (Create Index, Create View, Analyze Table)



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1. Design database untuk suatu Perusahaan

a. Tabel Departmen

Query :

```
CREATE TABLE department (  
    department_id NUMBER PRIMARY KEY,  
    department_name VARCHAR2(50) NOT NULL,  
    manager_id NUMBER,  
    location_id NUMBER  
);
```

Hasil Query :

Name	Null?	Type
DEPARTMENT_ID	NOT NULL	NUMBER
DEPARTMENT_NAME	NOT NULL	VARCHAR2(50)
MANAGER_ID		NUMBER
LOCATION_ID		NUMBER

b. Tabel Employees

Query :

```
CREATE TABLE employee (  
    employee_id NUMBER PRIMARY KEY,  
    last_name VARCHAR2(50) NOT NULL,  
    email VARCHAR2(50),  
    salary NUMBER(10,2),  
    commission_pct NUMBER(4,2),  
    hire_date DATE NOT NULL,  
    department_id NUMBER,  
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)  
);
```

Hasil Query :

Name	Null?	Type
EMPLOYEE_ID	NOT NULL	NUMBER
LAST_NAME	NOT NULL	VARCHAR2(50)
EMAIL		VARCHAR2(50)
SALARY		NUMBER(10,2)
COMMISSION_PCT		NUMBER(4,2)
HIRE_DATE	NOT NULL	DATE
DEPARTMENT_ID		NUMBER

c. Tabel Suppliers

Query :

```
CREATE TABLE supplier (  
    supplier_id NUMBER PRIMARY KEY,  
    supplier_name VARCHAR2(100) NOT NULL,  
    contact_email VARCHAR2(50)  
);
```

Hasil Query :

Name	Null?	Type
SUPPLIER_ID	NOT NULL	NUMBER
SUPPLIER_NAME	NOT NULL	VARCHAR2(100)
CONTACT_EMAIL		VARCHAR2(50)

Tabel products

Query :

```
CREATE TABLE product (  
    product_id NUMBER PRIMARY KEY,  
    product_name VARCHAR2(100) NOT NULL,  
    stock_quantity NUMBER DEFAULT 0,  
    supplier_id NUMBER,  
    CONSTRAINT fk_prod_supp FOREIGN KEY (supplier_id) REFERENCES supplier(supplier_id)  
);
```

Hasil Query :

Name	Null?	Type
PRODUCT_ID	NOT NULL	NUMBER
PRODUCT_NAME	NOT NULL	VARCHAR2(100)
STOCK_QUANTITY		NUMBER
SUPPLIER_ID		NUMBER

2. Analisis Table Menggunakan Compute Statistic

Query :

```
--2  
ANALYZE TABLE departments COMPUTE STATISTICS;  
ANALYZE TABLE employees COMPUTE STATISTICS;
```

Hasil Query :

Table DEPARTMENTS analyzed.

Table EMPLOYEES analyzed.

3. Isi data untuk table

Table Departments

Query :

```
INSERT ALL
  INTO department VALUES (10, 'Administration', 200, 1700)
  INTO department VALUES (20, 'Marketing', 201, 1800)
  INTO department VALUES (30, 'Purchasing', 114, 1700)
  INTO department VALUES (40, 'Human Resources', 203, 2400)
  INTO department VALUES (50, 'Shipping', 121, 1500)
  INTO department VALUES (60, 'IT', 103, 1400)
  INTO department VALUES (70, 'Public Relations', 204, 2700)
  INTO department VALUES (80, 'Sales', 145, 2500) -- Penting untuk View empvu80
  INTO department VALUES (90, 'Executive', 100, 1700)
  INTO department VALUES (100, 'Finance', 108, 1700)
SELECT 1 FROM DUAL;
```

Hasil Query :

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	114	1700
4	40	Human Resources	203	2400
5	50	Shipping	121	1500
6	60	IT	103	1400
7	70	Public Relations	204	2700
8	80	Sales	145	2500
9	90	Executive	100	1700
10	100	Finance	108	1700

Tabel Employees

Query :

```

INSERT ALL
INTO employee VALUES (101, 'Smith', 'smith@mail.com', 4000, NULL, TO_DATE('15-JAN-2020', 'DD-MON-YYYY'), 10)
INTO employee VALUES (102, 'Johnson', 'johnson@mail.com', 5000, NULL, TO_DATE('10-FEB-2021', 'DD-MON-YYYY'), 20)
INTO employee VALUES (103, 'Williams', 'williams@mail.com', 4500, NULL, TO_DATE('05-MAR-2019', 'DD-MON-YYYY'), 30)
INTO employee VALUES (104, 'Jones', 'jones@mail.com', 6000, NULL, TO_DATE('20-APR-2022', 'DD-MON-YYYY'), 40)
INTO employee VALUES (105, 'Brown', 'brown@mail.com', 3500, NULL, TO_DATE('12-MAY-2023', 'DD-MON-YYYY'), 50)
INTO employee VALUES (106, 'Davis', 'davis@mail.com', 7000, 0.1, TO_DATE('01-JUN-2018', 'DD-MON-YYYY'), 80) -- Dept 80
INTO employee VALUES (107, 'Miller', 'miller@mail.com', 7500, 0.15, TO_DATE('18-JUL-2017', 'DD-MON-YYYY'), 80) -- Dept 80
INTO employee VALUES (108, 'Wilson', 'wilson@mail.com', 7200, 0.1, TO_DATE('22-AUG-2019', 'DD-MON-YYYY'), 80) -- Dept 80
INTO employee VALUES (109, 'Moore', 'moore@mail.com', 9000, NULL, TO_DATE('30-SEP-2015', 'DD-MON-YYYY'), 90)
INTO employee VALUES (110, 'Taylor', 'taylor@mail.com', 8500, NULL, TO_DATE('14-OCT-2016', 'DD-MON-YYYY'), 100)
SELECT 1 FROM DUAL;

```

Hasil Query :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSIO...	HIRE_DATE	DEPARTMENT_ID
1	101	Smith	smith@mail.com	4000	(null)	15-01-2020	10
2	102	Johnson	johnson@mail.com	5000	(null)	10-02-2021	20
3	103	Williams	williams@mail.com	4500	(null)	05-03-2019	30
4	104	Jones	jones@mail.com	6000	(null)	20-04-2022	40
5	105	Brown	brown@mail.com	3500	(null)	12-05-2023	50
6	106	Davis	davis@mail.com	7000	0,1	01-06-2018	80
7	107	Miller	miller@mail.com	7500	0,15	18-07-2017	80
8	108	Wilson	wilson@mail.com	7200	0,1	22-08-2019	80
9	109	Moore	moore@mail.com	9000	(null)	30-09-2015	90
10	110	Taylor	taylor@mail.com	8500	(null)	14-10-2016	100

4. Analisis Table data setelah di isi data

Query :

```
ANALYZE TABLE departments COMPUTE STATISTICS;  
ANALYZE TABLE employees COMPUTE STATISTICS;
```

Hasil Query :

```
Table DEPARTMENTS analyzed.  
  
Table EMPLOYEES analyzed.
```

5. View Empvu80

Query :

```
CREATE OR REPLACE VIEW empvu80 AS  
SELECT employee_id AS id_number,  
       last_name AS name,  
       salary,  
       department_id  
FROM employee  
WHERE department_id = 80;
```

Hasil Query :

```
View EMPVU80 created.
```


6. Tampilkan empvu80

Query :

```
DESC empvu80;
```

Hasil Query :

Name	Null?	Type
-----	-----	-----
ID_NUMBER	NOT NULL	NUMBER(38)
NAME	NOT NULL	VARCHAR2(50)
SALARY		NUMBER(10,2)
DEPARTMENT_ID		NUMBER(38)

7. Non unique-index pada kolom FK pada table employee

Query :

```
CREATE INDEX emp_dept_idx  
ON employees(department_id);
```

Hasil Query :

```
Index EMP_DEPT_IDX created.
```

Link video : [LINK VIDEO](#)